

Understanding UCLA Student Attitudes toward Sustainable Fashion Consumption

Group 1

Alina Beaman 705989602

Valérie Ding 405953176

Alice Stessman 106138072

Abstract

This project investigates UCLA student attitudes and behavior surrounding sustainable choices with a focus on fashion. The aim is to understand how demographics relate to environmental sentiment and consumption habits. The study addresses a broader problem of the fashion industry's immense environmental toll and the increasing interest of consumers in sustainability amid concerns of greenwashing, overconsumption, and social justice. To explore this issue, we designed and distributed a questionnaire at UCLA, collecting 129 voluntary responses across various demographics and attitudes. Using Likert-scale responses on environmental views, we applied k-means clustering to segment students into groups of stronger or weaker sentiment regarding sustainable fashion. We then analyzed demographic patterns with those clusters and ran simple linear regression using demographic values to test for predictive power on environmental sentiment.

Our findings revealed that students who identified as politically liberal or as California residents were more likely to express concern for sustainability. Other demographics, such as race and religion, did not have clear correlations, but demonstrated notable patterns. We found that the majority of students maintain positive attitudes toward sustainability and environmental care, but there was a significant disconnect between beliefs and behavior. This was evident in the same majority purchasing clothes frequently and from first-hand sources. Additionally, students tend to consider style and price of clothing more than company values when shopping, indicating that aesthetic and economic considerations outweigh ethical concerns in practice. Our voluntary sample type limits the generalizability of our results, but these findings, while limited, highlight

how complex individual sustainable behavior can be in a culture driven by consumption. It indicates that awareness alone does not translate into sustainable action and that targeted education, policy, or cultural shifts may need to be enacted to bridge the gap between belief and practice.

Introduction

The fashion industry – only increasing in its trajectory – produces about 92 million tons of textile waste every year and exacerbates water pollution, microplastic contamination, and greenhouse emissions. It also takes immense advantage of extremely unethical labour, often underpaying workers in dangerous, neglectful conditions. Increased attention towards labour issues, sustainability, and social justice, especially through widespread media focus, is beginning to contribute to consumer psychographics that drive consumer behaviour. Crucially, attention towards what individuals can do on their own to create positive change is growing: boycott lists are able to spread more quickly than ever, explicit information on supply chains for consumer goods is more accessible than ever, and companies market sustainability and ethical labour more than ever. Consumers become skeptical of companies that do *and* companies that don't sell themselves as green or blue, wondering if they are earnest in their efforts or tunnel-visioned with profits. Though giant megacorporations are overwhelmingly at fault for the amount of pollution that makes the industry what it is today, consumer behavior is extremely valuable to analyze: inseparable from that pollution in question, for example, are the rising overconsumption patterns that were particularly worsened since the onset of the COVID-19 pandemic. The attention to aesthetic appeal, social media, marketability, self-commodification, and microtrends drives much of the purchasing behavior, but also drives much of what comes afterward: tons of clothes in landfills, cheaply-made clothes falling apart quickly, and a deeply unsustainable model for consumption.

Consumers begin to thrift, boycott, flip, and generally abstain, but they also have Instagram posts to put out, collections to upkeep, and a nonstop feed to follow – especially college students. With young people reliably being one of the largest demographic groups of fashion consumers and with LA being the hub that it is for modernity, pop culture influence, media attention, and fast-moving fashion, this study examines the UCLA-specific psychographic climate and analyzes campus attitudes towards sustainability in fashion. This study will apply methods and strategies previously employed by existing literature to a UCLA sample, making use of questionnaire data and clustering techniques to characterize the demographic.

Related Works

While research on consumer behavior is unending with the motivation of profit and research on sustainable fashion practices and consumption is growing with the anxiety of climate change, much of both kinds focus on broader populations without considering such a specific community. Our work builds on existing localized psychographic research to most effectively work with the data collected from the UCLA-wide distributed questionnaire.

One 2023 study in India examined demographic consumer behavior grouped by gender, age, income group, and quality of life, finding that quality of life and gender were particularly influential in identifying relationships to consumption and environmental care. Then, a qualitative 2014 study, after performing thematic analysis on focus group data, identified inconvenience, stigma, and other reservations as significant barriers to more sustainable purchasing behavior: informed by previous research examining the relationship between environmental knowledge and sustainable behavior, the study constructed the participants' narrative about self-efficacy, social and institutional influences, and prioritization of individual goals compared to collective goals.

Other questionnaires – a 2023 case study in India evaluating consumer perspectives, as well as a 2022 study on Chinese young adults examining skepticism about greenwashing and perceived efficacy of changing consumer habits and addressing the rationale of both sustainable and unsustainable consumption from the lens of risk theory – involve in-depth, targeted questions to better quantify the correlations between opinions.

Finally, of the most relevant, we refer to a 2020 review of 97 studies that deconstructs green purchasing behavior through a multi-step analysis based on consumer theory and converges on individual factors, product attributes and marketing, and social factors. Its individual factors are divided into psychological, lifestyle, and sociodemographics, product attributes and marketing into inherent attributes and marketing, and social influence into social norms and social capital – all valuable in strategizing the direction of our questions developed for the distributed survey.

There is limited research examining college-related demographics like field of study, club involvement, and year and how they impact sustainable fashion habits, particularly at a large, diverse, and spendy institution like UCLA. By focusing on UCLA students, our research addresses a gap in understanding the sustainability behaviors and attitudes of young adults in a specific and influential age group.

Data and Methods

Following a similar questionnaire as the 2023 case study in India, we created a survey that includes demographic features such as gender, race, religion, political affiliation, residency, year in school, major, and extracurriculars. The rest of the survey consists of questions evaluating the respondents' views

toward sustainable fashion as well as consumption patterns. We have 16 likert questions where the respondent is presented with a statement, for example, “I think it is worth it for the individual consumer to make efforts to preserve and improve the environment”, and must answer to the extent of which they agree with the statement (strongly disagree, disagree, neutral, agree, or strongly agree). Finally there are four questions asking about the respondents’ priorities when buying clothing, how often they shop and where from, as well as what factors would encourage them to reduce their consumption or purchase more sustainable apparel.

We collected 129 data points over the course of two months, asking people from clubs at UCLA, friends, and classmates to fill out the survey. Because our data was not randomly collected, our results cannot be generalized to the entire UCLA population and can be seen as general trends and observations. This was also not an experiment so causation cannot be inferred.

The first data method we used is k-means clustering. We used the responses to the likert questions as our variables to cluster (higher number (from one to five) = higher positive responses to sustainable fashion and protecting the environment). In R, we used the NB Clust library and `fviz_nbclust()` function to find the optimal number of clusters (two). We then use the `kmeans()` function to form the cluster groups. These two clusters can be broken apart into group one that tended to have responses showing less care for the environment and group two that had a more positive sentiment toward sustainable fashion. For each demographic we visualized the proportion of each category that belongs to cluster one and the proportion that belongs to cluster two in order to find any patterns in how demographics correlate to a lack of care or care for the environment.

Next, to test potential predictive power we used simple linear regression with the response variable being the average of all likert responses and demographics as the explanatory variables. We made a separate model for each demographic variable to see if any particular category in a demographic variable had significant influence on one’s sentiment about the environment. For each model we plot the residuals to make sure our assumption of normality was met.

Finally, we made a few plots showing simple summary statistics for the consumption pattern questions to get an overall understanding of our sample population.

Results



Figure 1: Student Clusters

The optimal two clusters are visualized above (figure 1). The red one on the left represents the group that tended to have a more negative sentiment toward sustainable fashion, while the blue group on the right is the cluster that had responses showing more care for the environment. The number of data points in group two is a decent amount higher than in group one, showing that in general our sample population had more of positive sentiment toward sustainable fashion. This could be due to bias from the specific people that filled out the survey or illustrate that students at a liberal college in california like UCLA tend to have a positive sentiment toward sustainable fashion.

With the chosen clusters, we made bar graphs for each demographic. The y-axis represents the proportion of the total respondents that answered with one of categories under the bars on the x-axis. The light blue represents the group of participants that belongs to cluster two, while the dark blue represents cluster one.

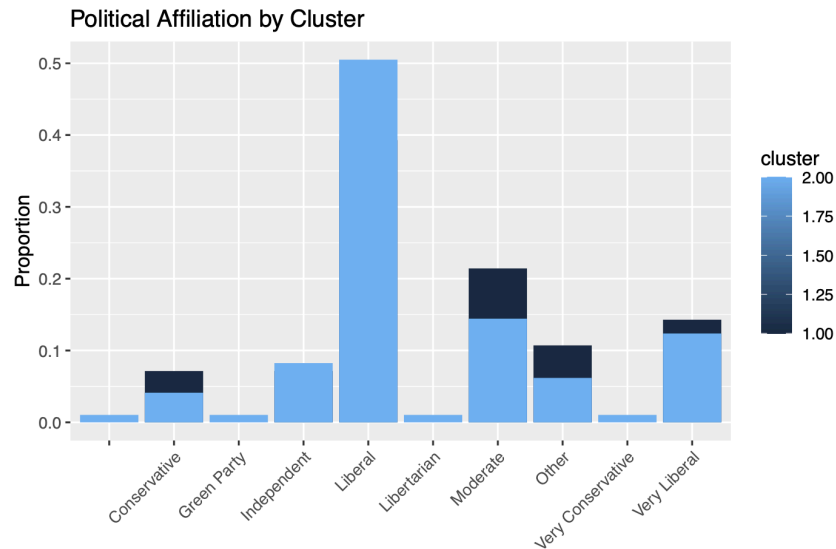


Figure 2: Political Affiliation by Cluster

Looking at political affiliation, it's very disproportional with half of the respondents identifying as liberal. The results here are what we might expect. In general the political left favors policies that focus on the environment, so it's no surprise that almost all who identified as liberal were categorized in cluster 2. Although the majority within categories conservative, moderate, and other, are in cluster 2, there is a decent amount that belong to cluster 1.

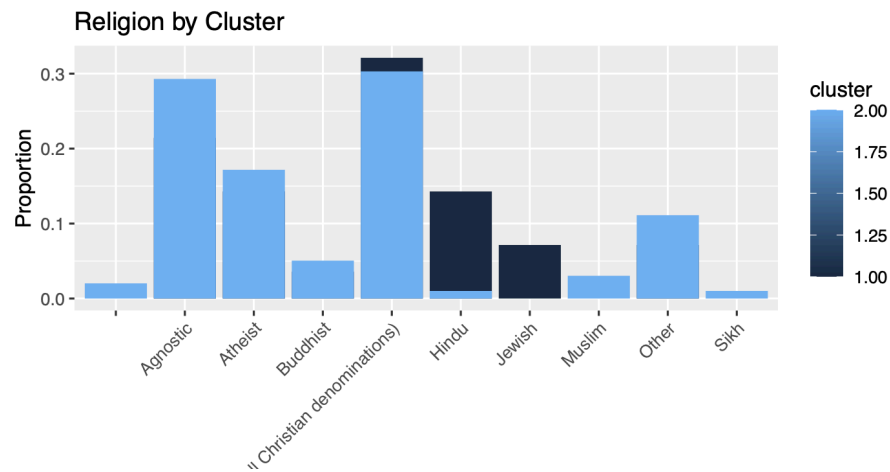


Figure 3: Religion by Cluster

Under the race demographic all the respondents that belonged to cluster one either identified as Asian or Pacific Islander or Black and under the religion demographic the majority of respondents that belonged to cluster one either identified as Hindu or Jewish (figure 3). Because of the small size of our sample size it's difficult to determine whether these are large trends but are interesting to note.

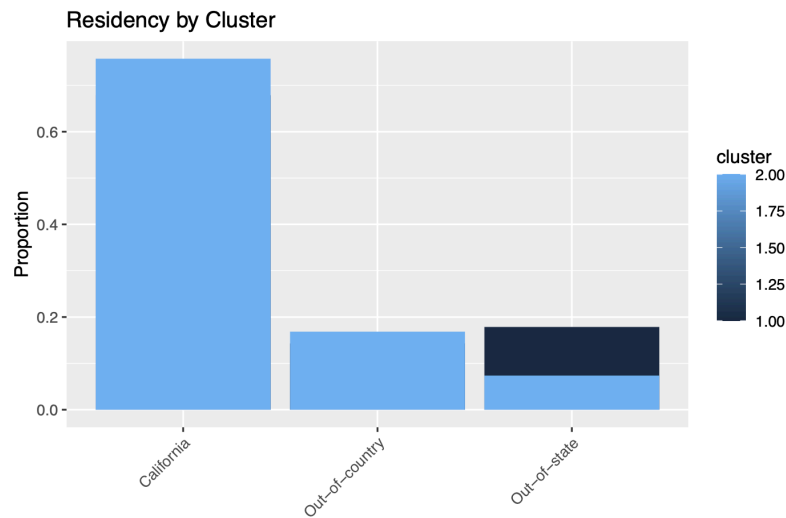


Figure 4: Residency by Cluster

Students were asked whether they are from California, out of the country, or out of the state. The respondents from California all belong to cluster 2, while all of the people that belong to cluster 1 are out-of-state. This illustrates how California's focus on the environment may have affected the sentiment of its residents. This may be related to politics as well since California is a blue state.

We analyzed the demographic variables race, religion, political affiliation, residency, and major using linear regression to test if any had predictive power on the average likert response of a respondent. The models followed some patterns seen from the clustering, but overall the variables had high p-values and were therefore not significant. This may be due to our small sample size and with a larger sample from the UCLA population we would expect significant results.

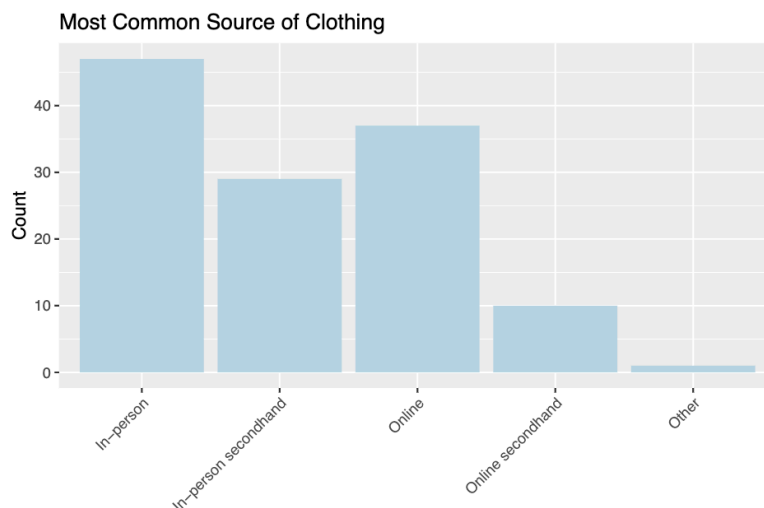


Figure 5: Most Common Source of Clothing

When asked how they most often shop, about 38% chose in person (new) and 30% online (figure 5). Despite most students responding with high positive sentiment toward sustainable fashion, the majority still buy first hand clothing as their main source of clothing. When asked how often they buy new clothing about 46% claimed once every few months, 38% once a month, 18% 1-2 times a year, 10% 3-4 times a month, and about 3% 5+ times a month. With many high likert responses, the majority of students surveyed would morally like to contribute to a world where the environment is taken care of, yet many still shop first-hand as their primary source of clothing and buy plenty of new clothes. There may be a dissonance between thoughts and actions for students today.

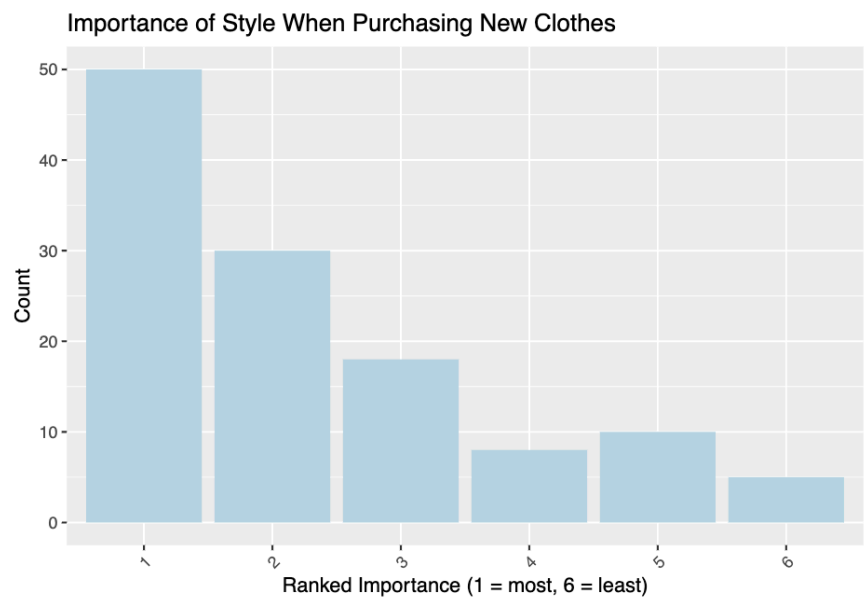


Figure 6: Importance of Style

We had respondents rank the following 6 factors from most important to least important when considering buying new clothing: style, practicality, material, company values, versatility, and price. When ranking style, about 40% placed style as the most important factor (figure 6).

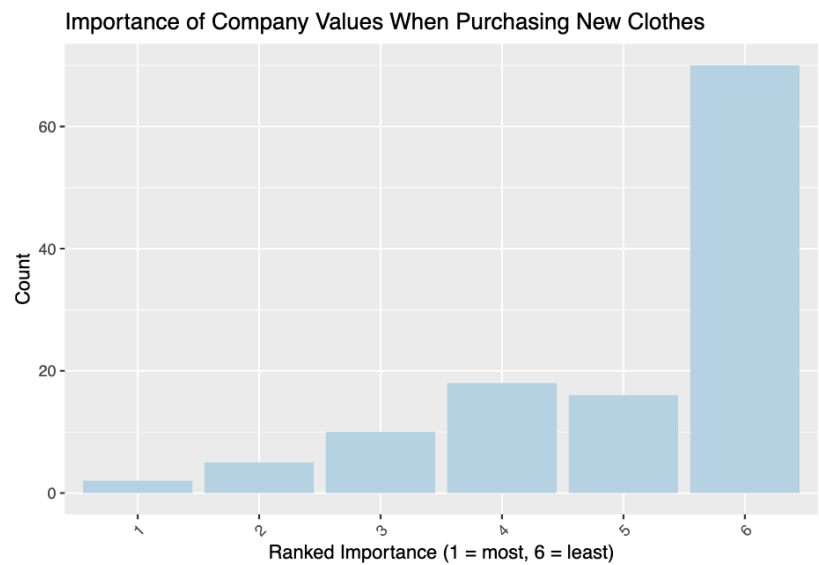


Figure 7: Importance of Company Values

When ranking company values, a majority placed it as the least important. Since this is a simple ranking this does not necessarily equate to not caring about the company values, but it illustrates that it is the least important compared to the other factors. This again may reveal that although some students may respond highly on the likert responses, they may not care as much to let it affect the way that they shop.

Discussion

Some of our findings confirmed our expectations, while others revealed intriguing observations. It was expected that students that identified with the political left and/or were from California would lean toward a positive view of sustainable fashion and we found evidence of that within our clusters. Some of the other demographic variables revealed certain religious and racial identities tended to belong to the group of students that showed less care for the environment, such as those identifying as Hindu or Jewish. We do not know, nor have a prediction of the underlying or correlated reason for this. Again, these findings are specific to our data as this was a volunteer survey.

Analyzing the consumer habits of our respondents, we found that there may be a dissonance between students' moral beliefs about the environment and their habits when relating to fashion. This is because a majority of students answered the likert questions with strong care for the environment, yet many still buy clothing frequently and from firsthand stores. It is difficult for people to change their shopping habits when they are trying to keep up with styles and are surrounded by constant consumerism in their everyday lives and in social media. Educating people about the negative effects of fast fashion is important, yet when some people are already aware, they need more than just that to push them to lower consumption.

Limitations and Future Work

The limits of our work are primarily caused by our sample: our fields of study, social circles, and academic interests, as well as the implied politics of even asking a question like ours are all factors that impact the generalizability of our results. Particularly, because of the absence of web scraping and our campus-specific demographic, our results and analyses are very limited, and we keep in mind that our sample is capped with those willing to engage with ideas of sustainability, political engagement, and conscious consumption. Our sample size is also very small, which is important to consider for the strength of the identified relationships between our variables. Our questionnaire design ran into some obstacles, including room for improvement with the question designs: for example, our questions for fields of study and extracurricular activities could be streamlined better.

Future work, especially considering the synthesis of ideas, methods, and theories in existing literature – i.e. consumer theory – can build significantly on our work by creating a greater body of data to examine, especially in determining which relationships are and are not generalizable to larger demographics. Our questions were intentionally designed to be minimally tedious for our audience, especially to try and collect as large of a sample as possible, but this is also responsible for a ceiling on nuance for our data. Focus groups, not unlike the aforementioned related work, among other qualitative studies, can also help contextualize a clearer infrastructure of opinions that produce our quantitative results. In general, our work can greatly benefit from connections to other quantitative work – greater in depth and breadth – and qualitative work – greater in narrative strength.

References

- Fullerton, S., & McCullough, T. (2023). Differences in eco-friendly attitudes and behaviors among consumers in the United States across an array of demographics. *Association of Marketing Theory and Practice Proceedings, 2023*, 42.
- Johnstone, M.-L., & Tan, L. P. (2015). Exploring the gap between consumers' green rhetoric and purchasing behaviour. *Journal of Business Ethics, 132*(2), 311–328.
- Lu, X., Sheng, T., Zhou, X., Shen, C., & Fang, B. (2022). How does young consumers' greenwashing perception impact their green purchase intention in the fast fashion industry? An analysis from the perspective of perceived risk theory. *Sustainability, 14*(20), 13473.
- Moitra, R., Rallan, P., & Bhalla, M. (2023). Demographic implications on sustainable consumption: An Indian case study. *Journal of Informatics Education and Research, 3*(2), 2618–2638.
- Sailer, A., Wilfing, H., & Straus, E. (2022). Greenwashing and bluewashing in Black Friday-related sustainable fashion marketing on Instagram. *Sustainability, 14*(3), 1494.
- Zhang, X., & Dong, F. (2020). Why do consumers make green purchase decisions? Insights from a systematic review. *International Journal of Environmental Research and Public Health, 17*(18), 6607.